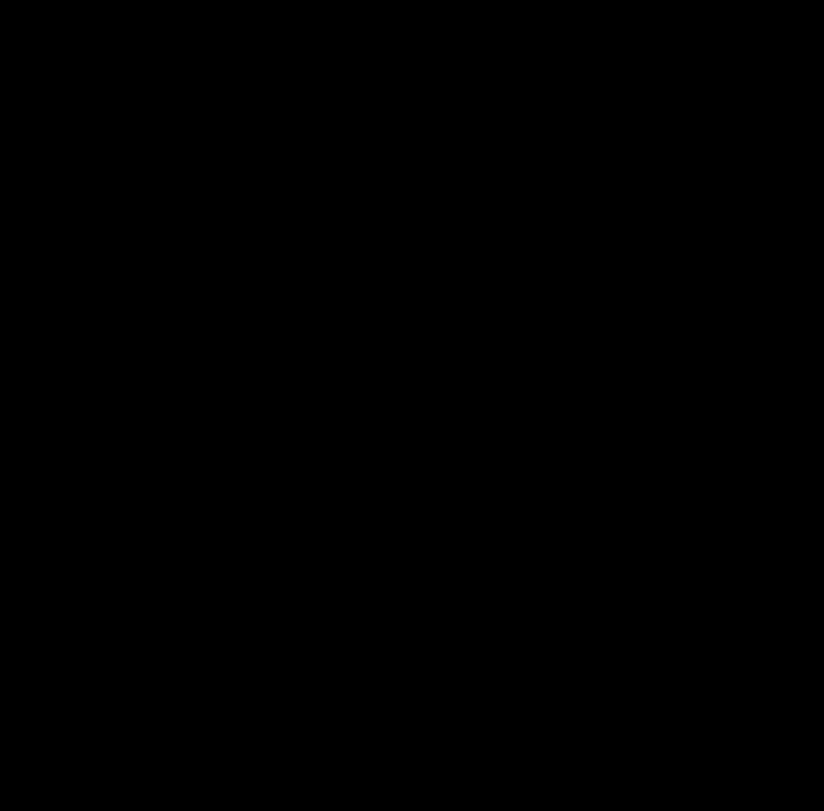




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TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

TRADE: SOFTWARE PROGRAMMING AND EMBEDDED SYSTEMS (SPE)

SUBJECT: SOFTWARE ENGINEERING

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATE:

1. This Exam paper is composed of **three sections (A, B and C)**. Follow the instructions given below and answer the indicated questions for a total of **100 marks**.

Section **A**: Fourteen (14) questions which are **Compulsory** **55 marks**

Section **B**: Among the five (5) questions, attempt any three (3) **30 marks**

Section **C**: Among the two (2) questions, attempt anyone (1) **15 marks**

2. Fill in your actual **names** and **Index number** on the provided space (Cover).
3. Do not remove or tear any page or part of this Questions and Answers Booklet.
4. Answer in the language in which the Examination is set.
5. In section **A**, answer questions in the provided space. In case of multiple-choice questions, just **circle** the letter corresponding to the correct option.
6. In section **B** and **C**, answer questions in the provided space after section C questions.
7. **Allowed materials:**
 - Blue or Black pen.
 - Geometrical Instruments
 - Silent non-programmable calculators

- 01.** A software development company decided to shift from traditional waterfall model to Agile methodology for better flexibility and faster delivery. The project manager informed the team that work would now be divided in short, manageable cycles rather than completing the entire project at once. This method would allow quicker feedback and adjustments throughout the development process. **(3marks)**

Fill-in-the blank Questions:

- a) In the Agile Model, development is done in, which are short cycles of work.
- b) These short cycle of development in Agile are called.....
- c) The purpose of using short cycle in Agile development is to deliver aworking product incrementally.

- 02.** At a software engineering seminar, participants were introduced to the OSLID principles; A fundamental set of guidelines for designing better object-oriented software. The trainer explained that each letter in SOLID represents a key principle that developers should remember to create maintainable and flexible system. The participants were asked basic questions to recall what each principle stands for and its main idea. **(4marks)**

Which of the following correctly describes the SOLID principles and their meanings?

(circle all that apply)

- a) The single Responsibility Principle states that a class should have only one reason to change.
- b) The Open/Closed Principle suggests that software entities should be open for extension but closed for modification.
- c) The Liskov Substitution mean objects of a superclass should be replaceable with objects of a subclass without affecting the program's correctness.
- d) The Interface Segregation Principle recommends that classes should depend on large, general-purpose interfaces rather than many specific ones.
- e) The Dependency Inversion Principle states that high-level modules should not depend on low-level modules; both should depend on abstractions.

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- 03.** During a software testing workshop, the instructor discussed several commonly used testing techniques to help ensure that software is working as expected. The instructor explained that some testing techniques focus on the functionality of the system (black-box testing), while others focus on the internal workings of the system (White-box testing). **(4marks)**

The trainees were asked to determine whether the following statements about software testing techniques were **True(T)** or **False(F)**

- a) Black-box testing focuses on testing the internal structure or implementation of the software system.....
- b) White-box testing requires knowledge of the system's internal workings, such as code structure and logic.....

- 04.** As the project kicks off, the project manager needs a document that defines the project's boundaries, specific goals, and the key individuals or groups involved in the project. This document will be shared with all stakeholders to ensure everyone is aligned with the project's scope and objectives. The manager asks the team to recall which document includes these details. **(4marks)**

Fill in the blanks using the correct terms.

- a) The document that outlines the
Objectives and participants of a project is called the Project Charter.
- b) The is the document that provides a detailed description of the project's goals, Participants and boundaries at the start of a project.

- 05.** At digital marketing seminar, the presenter explained different monetization strategies used by businesses to generate revenue from their products or services. One of the strategies discussed involved charging users on a recurring basis, Such as through subscriptions or memberships. The presenter asked the audience to identify monetization strategies typically involve recurring charge. **(4marks)**

Which of the following monetization strategies typically involves charging users on a recurring basis?

(circle all that apply)

- a) Subscription-based model
- b) One-time payment model
- c) Freemium model with recurring premium upgrades
- d) Pay-per-use model.

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- 06.** A software company is aiming to release, its latest application to users worldwide. The development team is discussing the concept of internationalization and its importance in their planning. **(4marks)**

Determine whether the following statements are **True(T)** or **False(F)**

Statement 1: The primary objective of internationalization in IT project is to ensure that the software's core functionality and design are adaptable to different languages and cultural requirements without needing significant re-engineering.....

Statement 2: Internationalization mainly involves translating the user interface into languages once the software is complete.
.....

- 07.** A software development company is considering the launch of a new project. During the initial stages of the project, the project manager discusses the importance of both technical feasibility and operational feasibility. The project team needs to evaluate whether the technology is capable of meeting the requirements (Technical feasibility) and if the project can be effectively implemented within the existing operational environment (operational feasibility). **(4marks)**

The manager asks the team to understand how these two types of feasibility are related and to identify the correct statements.

Which of the following statements accurately describe the relationship between technical feasibility and operational feasibility?

(circle all that apply)

- a) Technical feasibility focuses on whether the proposed technology and infrastructure can support the project, while operational feasibility assesses whether the project aligns with existing operational workflows and feasibility.
- b) Technical feasibility and operational feasibility are independent, with no impact on each other in project planning.
- c) Technical Feasibility ensures that the required software or hardware can be implemented, while operation feasibility ensure that the solution can be integrated and used effectively within the current organizational processes.
- d) Operational feasibility is more important than technical feasibility, as it determines whether the solution can be maintained in the long term.

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- 08.** A team of software developers is working on a new module for their application. They are discussing how to design the classes within this module to ensure it's maintainable, reusable, and easy to understand. They are considering the principles of cohesion and coupling in their design decisions. **(5marks)**

Match the descriptions on the left with the most appropriate consequence or characteristic on the right, based on the principles of high cohesion and low coupling:

Descriptions	Consequences/ Characteristics	Answer
1. A class has a single, well-defined responsibility.	A. High Coupling	1 =
2. Changes in one class are less likely to require changes in other classes.	B. Low Cohesion	2 = ...
3. A class performs a variety of unrelated tasks.	C. High Cohesion	3 = ...
4. A Classes is rely heavily on internal details of other classes	D. Reduced Maintenance Effort	4 =
5. A module is easy to understand, test, and reuse.	E. Increased Complexity and Fragility	5 =

- 09.** A software company has received a contract to build a mobile app for online banking. Before starting development, the project manager leads the team through the planning stage. They hold meetings to define project scope, estimate time and costs, allocate resources, and set communication guidelines. **(4marks)**

The project manager emphasizes that proper planning is essential to ensure that the project is completed on time, within budget, and meets client expectation.

Which of the following goals are most important during the planning stage of this project?

(Circle all that apply.)

- a) Defining the scope of work clearly
- b) Estimating the required time and resources
- c) Beginning coding to speed up delivery.
- d) Identifying potential risks and creating mitigation plans
- e) Setting clear communication channel and responsibilities.

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10. A software company has developed a new application that has been internationalized. They are now preparing to release it in several different countries. The project manager is explaining to the team why the subsequent step of localization is crucial for the success of their global rollout. **(5marks)**

Match the reasons on the left with the corresponding aspect of localization on the right that addresses that reasons:

Reasons why Localization is Essential	Aspects of Localization	Answer
1. Users prefer interacting with software in their native language.	A. Culture Adaption	1 = ...
2. Software needs to adhere to region-specific date, time, and number formats	B. Linguistic Translation	2 = ...
3. Certain symbols, image and color can have different meanings or be offensive in different cultures.	C. Format and Convention Adjustment	3 = ...
4. Legal and regulatory requirements can vary significantly between countries	D. Legal and Regulatory Compliance.	4 = ...
5. Units of measurement (e.g. currency, temperature) need to be appropriate for the local market.	E. Unit and Measurement Conversion	5 = ...

11. You are hired to lead a project for an e-commerce platform. The platform must be released quickly, but must also allow scaling and adding complex features later. The customer expects visible progress weekly, but some critical payment modules require high reliability. **(3marks)**

Which SDLC model(s) or practice(s) would you create a hybrid approach with to best meet these needs?

(Circle all that apply.)

- a) Start with Waterfall for the payment module; use Agile for the rest
- b) Implement Scrum sprints for frequent feature delivery
- c) Use only the V-Model to ensure quality from the start
- d) Combine DevOps practices to automate testing and deployment

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- 12.** Your team is working on a web-based e-commerce platform where users can browse products, add them to their cart, and make purchases. You are tasked with automating the entire end-to-end process using Selenium WebDriver for UI testing, TestNG for organizing tests, and integrating everything into Jenkins for continuous integration. **(4marks)**

Question:

To ensure that tests run smoothly and are scalable as the platform grows, you decide to design the automation strategy.

Which steps would you take to create a reliable and efficient testing workflow?

(Circle all that apply)

- a) Use TestNG to categorize the tests into smoke, regression, and performance, running only relevant tests based on the current build.
- b) Configure Selenium Grid to run tests on multiple browsers simultaneously and scale the test executions across different environments.
- c) Add Jenkins post-build actions to notify the team only if tests fail, skipping the results if tests pass.
- d) Automate the creation of test scripts in Selenium by generating the code using a recorder tool, avoiding writing any test cases manually.

- 13.** You are part of a development team building an online social media platform. The platform allows users to share posts, comment, and interact with others. As part of the development, you need to consider the ethical implications of collecting and storing user data, including personal information, browsing history, and interactions. **(4marks)**

Fill in the blank space of the following statement with the terms given below:

- a) European Union
- b) Role-Based
- c) Consent
- d) Delete

To ensure that the platform respects user privacy, you decide to implement a system where user data is only collected if the user has given explicit The data will be stored in a way that ensures only authorized personnel can access it, using access control mechanisms. Additionally, you will inform users of how long their data will be

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stored and provide them with the option to
their data at any time. You will also make sure that the data storage
complies with laws such as the GDPR.

- 14.** Given the current user base and product, which of the following monetization strategies would be most effective in maximizing revenue without sacrificing user experience? **(3marks)**

(Circle all that apply)

- a) Introduce a freemium model where users can access basic features for free but need to pay for more advanced tools and personalized advice.
- b) Implement in-app advertisements for free users, with the option for users to pay to remove the ads.
- c) Offer one-time purchases for specific wellness reports or insights, with no subscription options.
- d) Introduce a subscription-based model that provides users with ongoing access to all premium content, including new features and updates.

Section B: Attempt any three (3) questions

(30 marks)

- 15.** MediTrack is developing an EHR system using modular architecture for Patient Demographics, Appointments, Billing, and Prescriptions. During integration testing, the team finds that changes in the "Patient Demographics" module (e.g., adding a new language field) require modifications in both the "Appointments" and "Billing" modules. Additionally, the "Appointments" module combines tasks like scheduling, reminders, and doctor availability in the same classes. **(10marks)**

Questions:

Based on the challenges observed at MediTrack:

- a) How does the "Patient Demographics" module show High Coupling with other modules? Provide examples.
- b) How does this High Coupling affect maintainability and extensibility?
- c) How does the "Appointments" module demonstrate Low Cohesion? Provide examples.
- d) How does this Low Cohesion affect the "Appointments" module's future changes?

- 16.** Imagine you're building a simple online shop with two parts: "Product Display" (shows items) and "Shopping Cart" (lets users add items). The programmers tested each part separately, checking if the "Product Display" shows product names and prices, and if the "Shopping Cart" can add up prices correctly. However, when a user adds a product from the "Product Display" to the "Shopping Cart," only the price appears, not the product name. **(10marks)**

Questions:

Based on the issues observed in the online shop:

- Why did testing the "Product Display" and "Shopping Cart" separately fail to identify the issue with the product name in the cart? What does this tell us about the focus of separate testing?
 - Why was testing the process of adding a product to the "Shopping Cart" important for finding the problem? What does this kind of testing focus on?
 - What might have gone wrong in how the "Product Display" and "Shopping Cart" parts were designed to cause this issue? What might the "Product Display" be sending to the "Shopping Cart" about the product?
 - If the team had waited until the end to test how the "Product Display" and "Shopping Cart" work together, what might have been the impact?
- 17.** You are an IT student tasked with designing a **Student Attendance Management System (SAMS)** for your school's capstone project. Your role as part of the **planning team** requires you to demonstrate advanced project management planning and visualization skills based on the data given in the following table. **(10marks)**

Activity Code	Activity Name	Duration (days)	Dependencies	Start Date	End Date
A	Requirement Gathering	3	-	01-May	03-May
B	System Design	4	A	04-May	07-May
C	Database Design	3	B	08-May	10-May
D	UI/UX Prototyping	4	B	08-May	11-May
E	Backend Development	6	C, D	12-May	17-May
F	Frontend Development	5	D	12-May	16-May
G	Integration of Modules	3	E, F	18-May	20-May
H	Testing and Debugging	5	G	21-May	25-May
I	User Feedback and Iteration	3	H	26-May	28-May
J	Final Presentation & Delivery	2	I	29-May	30-May

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- a) Suggest any two (2) possible milestones for this project.
- b) Create Gantt chart for the project activities given in the table above.

18. Imagine you're leading a small team in Nyamata, Rwanda, with just **(10marks)** one week to understand the needs of local farmers for a new mobile application that provides weather updates and market prices. You have two options for gathering information:

- a) One-on-One Interviews (Stakeholder Interviews): Visit individual farmers in their fields or homes for detailed conversations. Each interview might take about 1-2 hours, including travel time.
- b) Group Discussions (Focus Groups): Organize meetings with a small group of farmers (5-8 people) to discuss their needs. Each focus group session might take 2-3 hours, including travel and setup.

Given the one-week deadline, you need to decide which data collection method would be more effective.

Questions:

Given the tight one-week deadline to understand the needs of local farmers in Nyamata for a weather and market price mobile application:

- a) Evaluate the effectiveness of one-on-one interviews for gathering in-depth information within this limited timeframe. What are the advantages and disadvantages of this approach considering the time constraint?
- b) Evaluate the effectiveness of group discussions (focus groups) for gathering a range of perspectives within the same timeframe. What are the advantages and disadvantages of this approach considering the time constraint?
- c) Based on your evaluation, which data collection method would you recommend your team prioritize for this one-week project in Nyamata? Justify your recommendation by comparing the strengths and weaknesses of both methods.
- d) What potential risks or limitations might arise from choosing focus groups due to the short timeframe, and how might you mitigate these risks within the one-week period?

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- 19.** A software project team is struggling due to high levels of conflict, frequent misunderstandings, and poor information flow between team members and stakeholders. As the project manager, you are tasked with identifying and evaluating which project management knowledge area needs the most improvement to resolve these challenges. **(10marks)**

Evaluate the relevance and potential impact of improving each of the following knowledge areas in addressing the current issues.

Options to evaluate:

- a) Cost Management
- b) Time Management
- c) Human Resource Management
- d) Communication Management

For each option, provide justification based on its role in managing team conflict and improving communication. Conclude by recommending the most appropriate knowledge area to prioritize, with reasons.

Section C: Attempt only one (1) question

(15 marks)

-
- 20.** Imagine a small cooperative of coffee farmers in the Kayonza district, wants to expand their local coffee sales by selling roasted coffee online to customers in Kigali and internationally. Currently, their manual process includes taking orders, checking stock, packaging, arranging transport, and paying members. They are looking for a digital system to manage these tasks more efficiently. **(15marks)**

Questions:

As a software design expert helping this coffee cooperative:

- a) Imagine you are building a basic digital system to manage online roasted coffee sales. Based on the current manual process, create a list of key "things" (entities) the system would need to track. For each, list important pieces of information (attributes) to store. Focus on what's essential for taking orders, managing stock, and processing payments.
- b) Now, using your list of "things" from part (a), design a simple visual representation (a logical diagram) showing how these "things" might relate. For example, how does a "Customer" relate to an "Order"? How does an "Order" relate to the "Roasted Coffee Stock"? Focus on the main connections that would make the online sales process work.

- 21.** The **City of Lumina** is funding a major **Community Learning Platform Software Development Project** to support digital education and online innovation. As an appointed **Software Project Manager and Architect**, you are responsible for creating a complete project schedule while dealing with uncertain task durations.

The following **activities** must be completed for the software platform to be developed and deployed successfully. For each task, you are given three-time estimates (in days):

Activity	Predecessor(S)	Optimistic (days)	Normal (days)	Pessimistic (days)
a) Obtain project Approval and Initial Funding	-	3	6	9
b) Finalize Software Requirements and Architecture Design	-	9	12	21
c) Procure Development Tools and Set Up Development Environment	-	6	7	12
d) Develop Core System Modules	A,B	13	16	19
e) Develop Integration Services and APIs	B,C	3	11	13
f) Conduct System Integration and Testing	E	4	6	8
g) Deploy the Platform to the Production Environment	F	2	5	8
h) Perform Final User Acceptance Testing (UAT) and Launch	G	2	4	6

As a technical and creative leader, you are required to **develop the full software project plan** using your project management and analytical skills.

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- Calculate and justify the expected time required to complete each activity using the formula for expected time estimation:
- Design a project activity network diagram based on your calculated expected durations and clearly demonstrate the critical path — the sequence of activities that will delay the entire project if delayed.
- Within your project network, which activities are immediate successors to both Activity A (Obtain Approval and Funding) and Activity B (Finalize Requirements and Architecture)?
- Identify and justify which activity acts as an immediate predecessor to Activity F (System Integration and Testing) in your network.

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